

Prose, mathematics, and Lean 4 code together

ScottLean4 · for Dana Scott

This template shows the three kinds of content you will interleave: ordinary **prose**, typeset **mathematics**, and blocks of **Lean 4 code**.

Prose flows as usual, with inline math such as $\forall n \in \mathbb{N}, n + 0 = n$, and displayed math:

$$\sum_{k=0}^n k = \frac{n(n+1)}{2}, \quad (a \wedge b) \rightarrow (b \wedge a).$$

A block of Lean 4 code is set verbatim; its Unicode ($\mathbb{N}$, $\rightarrow$, $\forall$) renders directly because we compile with XeLaTeX and a Unicode monospace font:

```
1 | theorem add_zero (n : ℕ) : n + 0 = n := by
2 |   simp
3 |
4 | theorem and_comm (a b : Prop) (h : a ∧ b) : b ∧ a :=
5 |   ⟨h.right, h.left⟩
```

And then prose resumes, and can refer back to the code above and to the display equation. The pattern — *paragraph, equation, code block, paragraph* — repeats for as long as the document needs.